Лекция 6

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Следствие. $f: U_\delta(x_0) \xrightarrow{\subset \mathbb{R}^k} Y \subset \mathbb{R}^l$

$f(\vec{x}_0) = \vec{y}_0 \in \text{int } Y$

$\vec{z} = \varphi(\vec{x}) \sim \vec{g}(\vec{f}(\vec{x}))$

или $\vec{z} = \vec{g}(\vec{y})$ $\quad d\vec{z} = \mathcal{D}_{\vec{g}}(\vec{y})\, d\vec{y}$

Док-во: 1) $\vec{z} = \vec{g}(\vec{y})$ $\quad d\vec{z} = \mathcal{D}_{\vec{g}}(\vec{y})\, d\vec{y}$

2) $\vec{z} = \vec{\varphi}(\vec{x})$ $\quad d\vec{z} \sim \mathcal{D}_{\varphi}(\vec{x})\, d\vec{x} =$

$= \mathcal{D}_{\vec{g}}(\vec{y}_0) \cdot \mathcal{D}_{\vec{f}}(\vec{x}_0)\, d\vec{x}$

Опр. $f: U_\delta(\vec{x}_0) \xrightarrow{\subset \mathbb{R}^k} \mathbb{R}$

$\frac{\partial f}{\partial x^k}(x)$ опр. в $U_\delta(\vec{x}_0)$

Тогда: $\frac{\partial}{\partial x^m}\left(\frac{\partial f}{\partial x^k}\right)(x_0)$ – частная производная второго порядка ф-ии f.

$\frac{\partial^2 f}{\partial x^m \partial x^k}(\vec{x}_0)\left(f_k(\vec{x}_0)\right)$

$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2}, & (x,y) \neq (0,0) \\ (x,y) = (0,0) \end{cases}$

$\frac{\partial f}{\partial x}(0,0) = 0$

$\frac{\partial f}{\partial x} = \frac{(y(x^2-y^2)+2x^2y)(x^2+y^2) - xy(x^2-y^2)2x}{(x^2+y^2)^2} =$

$= \frac{(3x^2y - y^3)(x^2+y^2) - 2x^2y(x^2-y^2)}{(x^2+y^2)^2} = \ldots =$

$= \frac{y(x^4+4x^2y^2-y^4)}{(x^2+y^2)^2}$

$\frac{\partial f}{\partial x}(x,y) = \begin{cases} \frac{y(x^4+4x^2y^2-y^4)}{(x^2+y^2)^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

$\frac{\partial f}{\partial x}(0,y) = -y$

$\frac{\partial^2 f}{\partial y \partial x}(0,0) = -1$

$\frac{\partial f}{\partial y}(x,y) = \begin{cases} \frac{-x(y^4+4x^2y^2-x^4)}{(x^2+y^2)^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

$\frac{\partial f}{\partial y}(x,0) = x$

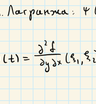
$\frac{\partial^2 f}{\partial x \partial y} = 1$

Ymb $f: U_{\delta_0}(x_0, y_0) \to \mathbb{R}$

$\frac{\partial^2 f}{\partial x \partial y}$ и $\frac{\partial^2 f}{\partial y \partial x}$ опр. в $U_{\delta_0}(x_0, y_0)$

и непр. в $\tau(x_0, y_0) \Rightarrow \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$

Док-во: $\delta = \frac{\delta_0}{\sqrt{2}}$ $\quad t \in (-\delta, \delta)$



$\omega(t) = f(x_0+t, y_0+t) - f(x_0+t, y_0) - f(x_0, y_0+t) + f(x_0, y_0)$

Рассмотрим $\varphi(x) = f(x, y_0+t) - f(x, y_0)$

$\varphi(x_0+t) - \varphi(x_0) = \varphi'(\xi_1)\, t \quad \xi_1 \in (x_0, x_0+t)$

$\omega(t) = \left(\frac{\partial f}{\partial x}(\xi_1, y_0+t) - \frac{\partial f}{\partial x}(\xi_1, y_0)\right) t$

$\psi(y) = \frac{\partial f}{\partial x}(\xi_1, y)$

т. Лагранжа: $\psi(y_0+t) - \psi(y_0) = \psi'(\eta_2) \cdot t$

$\eta_2 \in (y_0, y_0+t)$

$\omega(t) = \frac{\partial^2 f}{\partial y \partial x}(\xi_1, \eta_2)\, t^2 \qquad \frac{\omega(t)}{t^2} = \frac{\partial^2 f}{\partial y \partial x}(\xi_1, \eta_2) \xrightarrow{t \to 0} \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$

$\lim_{t \to 0} \frac{\omega(t)}{t^2} = \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$

$\gamma(y) = f(x_0+t, y) - f(x_0, y)$

$\exists \eta_2 : \omega(t) = \left(\frac{\partial f}{\partial y}(x_0+t, \eta_2) - \frac{\partial f}{\partial y}(x_0, \eta_2)\right) t$

$\exists \eta_1 \in (x_0, x_0+t) : \omega(t) = \frac{\partial^2 f}{\partial x \partial y}(\eta_1, \eta_2)\, t^2$

$\frac{\omega(t)}{t^2} = \frac{\partial^2 f}{\partial x \partial y}(\eta_1, \eta_2) \xrightarrow{t \to 0} \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0)$

$\lim_{t \to 0} \frac{\omega(t)}{t^2} = \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0)$

Опр. $f: U_\delta^{\subset \mathbb{R}^k}(\vec{x}_0) \to \mathbb{R}$

Все частные производные порядка $(m-1)$ определены в этой же окрестности $U_\delta(\vec{x}_0)$ и дифференцируемы в x_0, тогда d^k раз диф. в т. $\vec{x}_0$

Аналогично $d^k f(\vec{x}_0) = d\left(d^{k-1} f(\vec{x}_0)\right)$

$f(x,y) = xy$

$df = x\,dy + y\,dx$

$d^2 f = 2\,dx\,dy$

Ymb. $f: U_\delta^{\subset \mathbb{R}^k}(\vec{x}_0) \to \mathbb{R}$

k раз дифф. в т. $\vec{x}_0 \Rightarrow$

$\Rightarrow d^k f(\vec{x}_0) = \sum_{i_1=1}^{n} \ldots \sum_{i_k=1}^{n} \left(\frac{\partial^k f}{\partial x^{i_1} \ldots \partial x^{i_k}}\, dx^{i_1} \ldots dx^{i_k}\right)$

Док-во: $df(\vec{x}_0) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}\, dx_i$

$d^2 f(\vec{x}_0) = d\left(\sum_{i=1}^{n} \frac{\partial f}{\partial x^i}\, dx_i\right) = \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 f}{\partial x^j \partial x^i}\, dx_j\, dx_i$

Опр. F° – ф-ции $f: \mathbb{R}^n \to \mathbb{R}$: имеют k раз дифф. в $\mathbb{R}^n$

$\lambda f \quad \forall \vec{x}_0 \subset \mathbb{R}^n \quad h(\vec{x}_0) = \lambda f(\vec{x}_0)$

$f+g \quad \forall \vec{x}_0 \subset \mathbb{R}^n \quad h(\vec{x}_0) = f(\vec{x}_0) + g(\vec{x}_0)$

F° – лин-во $f: \mathbb{R}^n \to \mathbb{R}$

Опр. $\hat{A}: X \to Y$ линейный оператор, если

$\forall f, g \in X \ \forall \lambda, \mu \in \mathbb{R} \hookrightarrow \hat{A}(\lambda f + \mu g) = \lambda \hat{A} f + \mu \hat{A} g$

$\frac{\partial}{\partial x^k} : F^k \to F^{k-1}$

$\frac{\partial f}{\partial x^k}$

Опр. $\hat{A}_1, \ldots, \hat{A}_n : X \to Y$ лин-ые операторы,

$\lambda_1, \ldots, \lambda_n \in \mathbb{R}$

$\hat{A} = \lambda_1 \hat{A}_1 + \ldots + \lambda_n \hat{A}_n$ $\hat{A}$ линейный оператор.

$f \in X \quad \hat{A}_1 f, \ldots, \hat{A}_n f \in Y$

$\lambda_1 \hat{A}_1 f : \lambda_1 \hat{A} f \in Y$

$\lambda_1 \hat{A}_1 f + \ldots + \lambda_n \hat{A}_n f \in Y$

$\lambda_1 \frac{\partial}{\partial x^1} + \ldots + \lambda_n \frac{\partial}{\partial x^n}$

$\lambda_1 \frac{\partial f}{\partial x^1} + \ldots + \lambda_n \frac{\partial f}{\partial x^n} = (\text{grad } f, l) \quad l = (\lambda_1, \ldots, \lambda_n)$

$dx^1 \cdot \frac{\partial}{\partial x^1} + \ldots + dx^n \frac{\partial}{\partial x^n}$

Опр. $\hat{A}: X \to Y$ $\quad \hat{A}, \hat{B}$ – линейные операторы

$\hat{B}: Y \to Z$ $\quad \hat{C}: X \to Z$ $\quad \hat{C} = \hat{B}\hat{A}$

композиция операторов $\hat{A}$ и $\hat{B}$

$\hat{C} f = \hat{B}(\hat{A} f)$

$\hat{A} = dx^1 \frac{\partial}{\partial x^1} + \ldots + dx^n \frac{\partial}{\partial x^n}$

$\hat{A}^2 f = d^2 f$